Prerana Puttaswamy

Long Beach, CA | +1 (562) 832-5577 | prerana22002@gmail.com | GitHub | LinkedIn

EDUCATION

California State University, Long Beach — May 2026
Master of Science in Computer Science — GPA: 3.5

Visvesvaraya Technological University, India — May 2024
Bachelor of Engineering in Computer Science — GPA: 3.8

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL
Backend: FastAPI, Flask, REST APIs, WebSockets, Microservices
Frontend: Next.js, HTML, CSS
Cloud & DevOps: AWS, Docker, Kubernetes, Git
Databases & Queues: MySQL, Redis
AI & ML: PyTorch, OpenCV, MediaPipe, Ollama (LLM/GenAI)
Concepts: Distributed Systems, Fault-Tolerant Architecture, Event-Driven Systems, OOP, Data Structures & Algorithms, Agile Methodologies
Certifications: AWS Cloud Technical Essentials | REST APIs with Flask – Coursera (IBM)

EXPERIENCE

Software Engineering Intern — Sept 2022 – May 2023
Hewlett Packard Enterprise, Bengaluru, India

- Served as a backend engineer on the infrastructure observability team; designed and deployed RESTful APIs using Python Flask for authentication, diagnostics, and data retrieval, improving backend reliability across the distributed cluster platform.
- Built BUDHA, a conversational AI assistant for querying Kubernetes resources, system logs, and cluster health data in natural language, streamlining incident investigation and reducing reliance on manual kubectl-based diagnostics.
- Engineered automated Kubernetes health monitoring workflows and structured logging pipelines across multiple microservices, improving observability and reducing time to identify and resolve production issues.

PROJECTS

KubeGPT Dashboard – Intelligent Kubernetes Debugging Tool [GitHub]
FastAPI, Kubernetes, Python, Ollama (LLM/GenAI), Docker

- Built an intelligent Kubernetes debugging assistant that automates diagnosis of pod failures (CrashLoopBackOff, ImagePullBackOff), reducing manual troubleshooting time by ~40–60%.
- Designed a hybrid inference pipeline combining deterministic rule-based analysis with LLM fallback via Ollama, reducing unnecessary AI calls by ~70% and improving response latency.
- Automated extraction of root cause, evidence, and fixes from unstructured logs, transforming raw Kubernetes output into actionable structured incident reports.

StreamMind AI – Real-Time Event Processing System [GitHub]
FastAPI, Redis, Celery, Next.js, WebSockets, Docker, MySQL

- Designed a fault-tolerant, event-driven architecture using Redis Streams and Celery workers to process 100+ events/min, with backend APIs handling event ingestion, analytics, and failure tracking across multiple event types.
- Replaced HTTP polling with persistent WebSocket connections, reducing UI update latency by ~80% and enabling real-time data streaming to the frontend.
- Developed anomaly detection logic that flagged system failures when error rates exceeded defined thresholds, improving observability and reducing undetected incidents.
- Containerized a multi-service architecture using Docker Compose, enabling one-command reproducible deployment across environments.

RESEARCH & ACADEMIC PROJECTS

Real-Time Sign Language Translation Using Deep Learning (2024) [Link] – Published peer-reviewed research on a bidirectional gesture recognition system; IARJSET Vol. 11, Issue 4. DOI: 10.17148/IARJSET.2024.11494

Leader Election Under Crash-Recovery Fault Models (2025) – Academic research evaluating Raft and Bully protocols under network partitions and packet loss, analyzing convergence latency and message complexity trade-offs in distributed systems.

Cache Optimization for Hybrid In-Memory Computing Systems (2024) – Academic research surveying cache management techniques for DRAM/NVM architectures, comparing latency, energy efficiency, and throughput trade-offs.